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RIFT in Dragonfly Topologies

Abstract

RIFT extensions for dragonfly topologies.

Requirements Language

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1. Introduction

RIFT today is standardized to deal with CLOS variant fabrics with some horizontal link exceptions. Given that interconnecting multiple CLOS via a dragonfly variant is an interesting topology (whether it's a full mesh or some kind of non-completely meshed regular lattice) this document addresses the resulting changes necessary to base RIFT specification to support dragonfly interconnected CLOS fabrics. The reader is advised that due to complexity of figures involved the ASCII version of the document leaves those out.

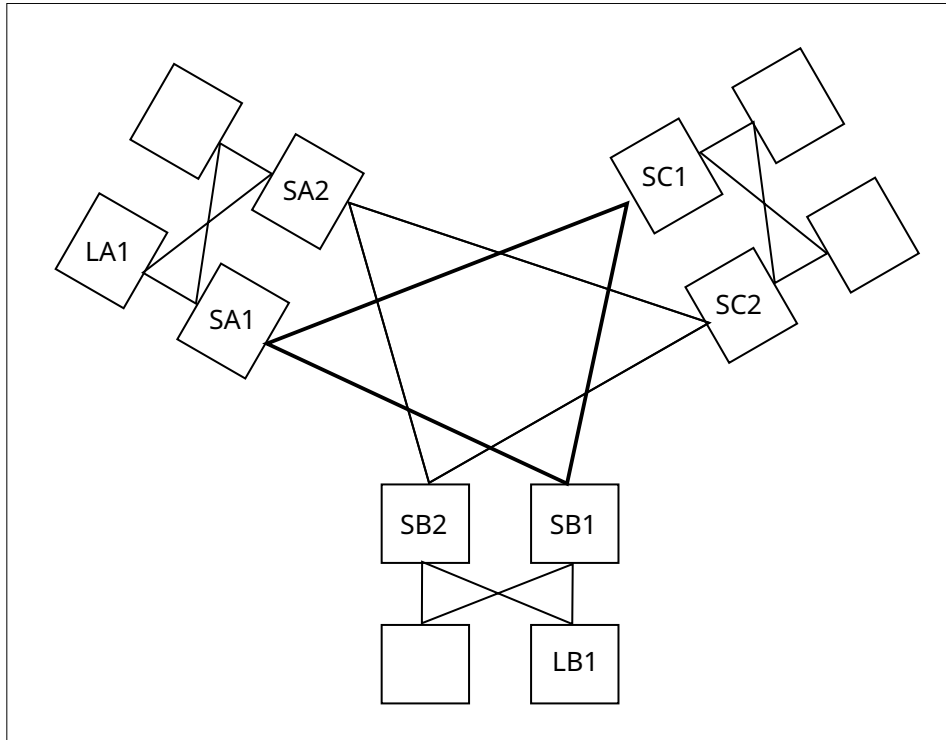


Figure 1: Sparse Dragonfly of CLOS Fabrics

To start with, [Figure 1](#) visualizes three simple single plane fabrics interconnected via a DragonFly+ backbone. The behavior of standard RIFT is better understood if we look at the homomorphic version of the same topology in [Figure 2](#). We can see that it is nothing else but a multi-plane CLOS with a lot of broken links for standard RIFT. The planes consist of S_{x_1} and S_{x_2} ToFs in each CLOS. Given this, leaf LB1 should be connected to SA1 to be in the plane and since it is not, SA1 will deduct that leaf LB1 fell off the plane 1 and negatively disaggregate it. Unfortunately the same is true for leaf LB1 from the view the SA2 in 2nd plane and it will negatively disaggregate it as well. Hence, leaf LA1 will not have any possibility to forward to LB1 using standard RIFT computed forwarding. This points us already to the first modification needed; we have to relax RIFT to forward through the horizontal links on ToFs and this will be the starting point of the next section.

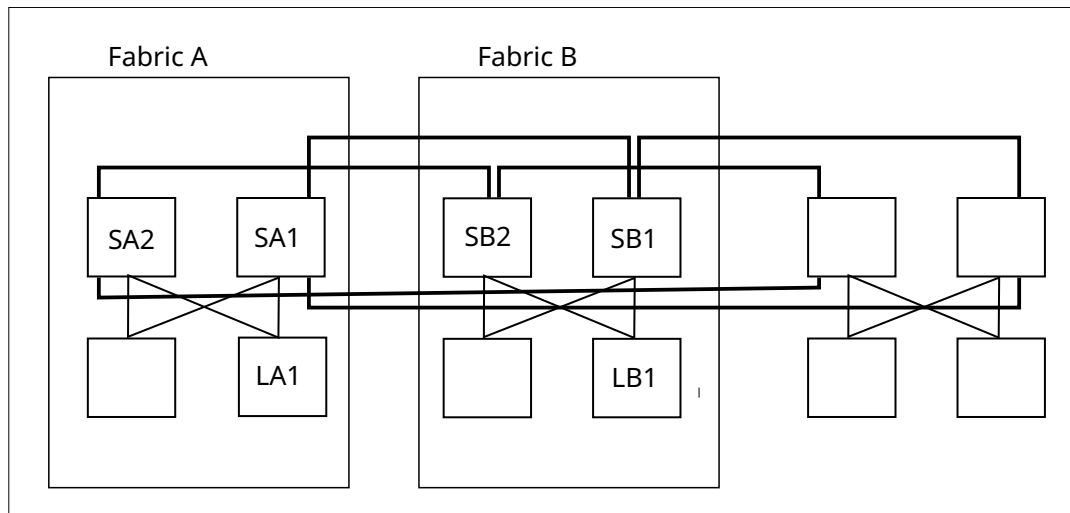


Figure 2: Homomorphic View of Sparse Dragonfly as a Multi-Plane CLOS

2. Glossary

The following terms are used in this document.

DF+ capable ToF:

ToF that provides DF+ extensions, both in recognizing the inter fabric links and computation procedures necessary to support those. The resulting combination allows the use of RIFT with dragonfly topologies overall.

Horizon:

We define horizon as a concept differentiating between inter fabric links and southbound pointing standard RIFT intra fabric links on a ToF. Both type of links need a different FIB to support alternate next hop when routing between fabrics.

Inter Fabric Planes or IF-planes:

Multi-Plane that spans multiple fabrics.

Inter Fabric link or IF links:

A horizontal ToF link between two fabrics.

Alternate Next Hop:

A next hop through an IF interface that does not represent the shortest path through the inter fabric horizon but necessitates the receiving node to use the next hop on the shortest path to the destination fabric (direct next hop).

3. Horizontal Link Behavior at ToF Level

Dragonfly+, being basically, when seen a single fabric, a multi-plane CLOS with many broken links (which we will call inter fabric planes or IF planes to distinguish them from multi-plane within a fabric later) will somehow need to change the behavior of RIFT to allow forwarding via horizontal links at ToF level lest we end up inverting the fabric and force leaves to deal with transit traffic. Moreover, the necessity to deal with new mis-cabling concepts leads us to change the solution framework and consider this configuration not as a single fabric but as a multi-fabric setup with dragonfly links building inter fabric planes now. Additionally we will have to allow adjacencies on ToF horizontal links to another fabric and permit those to forward through such inter fabric planes while distinguishing such inter fabric (or IF) links from normal horizontal ToF "multi-plane ringing". Hence in [Figure 2](#) instead of the first assumption of a single fabric we break out fabric A and fabric B and consider the links SA2-SB2 and SA1-SB1 as two "inter fabric DF+" links, or in short, as already introduced, IF links. And fortunately enough, IF links, just like all other horizontal ToF links, are considered northbound from both sides and northbound flooding rules apply, an ideal thing since with that ToFs will see full topology of their inter fabric plane.

RIFT used in such DF+ configuration will require on ToF not only a DF+ capability flag but a fabric ID now which has to be distinct in each of the CLOS. In case of non-DF+ mode a ToF will declare such links miscabled, once enabled to operate in DF+ it will mark those links as IF links. Given `fabric\_id` is an optional schema element a ToF operating in DF+ mode will reject all links to other ToFs without `fabric\_id` value set or not indicating DF+ mode as mis-cabled to prevent a mixture of non-DF+ and DF+ ToFs in a setup. On the other hand, a ToF indicating DF+ capability and showing matching fabric id is clearly a normal horizontal multi plane ring in the same fabric.

4. First Route Computation Change

Now that we can detect IF links reliably we can also remove those from the computations used in negative disaggregation as first step. This will prevent ToFs in fabric A negatively disaggregating Fabric B prefixes, a desirable behavior. Not being able to forward from Fabric A to fabric B is obviously a far less desirable behavior and hence a ToF in DF+ mode needs to extend its route computation by a special southbound DF+ computation where we use SPF taking in first step all IF links and the nodes behind them as candidates. This computation will result in a "direct inter fabric forwarding database" containing amongst others shortest path to prefixes in fabric B or in other words, direct inter fabric next hops. [Section 5](#) will expound further how that database is used.

4.1. Additional Bi-Sectional Bandwidth Route Computation Change

One of the DF+ properties is that it not only provides a direct path to a destination but guarantees that destinations are reachable via additional, alternate next hop to increase the bi-sectional bandwidth. In our example SB1 forwarding to LA1 can take instead of SA1 directly a path

through SC1 relying on it forwarding to SA1. To support this we introduce an additional SPF computation which takes in first 2 iterations only IF links and generates a "indirect inter fabric forwarding database". [Section 5](#) will expound further how that database is used.

Computing such alternate next hops will have the other beneficial effect of actually providing a backup path in case the direct IF plane link to another fabric becomes unavailable.

4.2. Dragonfly with Multi-Plane CLOS Fabrics

Most complex case of RIFT deployment would be a dragonfly topology of CLOS fabrics which are in themselves already multi-plane fabrics. To present it as homomorphic graph [Figure 3](#) is included. The symmetry is obvious, we end up with the normal RIFT ringing within the fabric, e.g. r_A for fabric A and then for the inter fabric planes dragonfly is basically the according ringing itself, here IR_1 and IR_2. Observe that the northbound flooding occurring on all those links will present each ToF with the full topology of the dragonfly, a necessary condition for proper disaggregation and further reachability computations. If the intra fabric ToF ringing should be avoided a tunnel between the ToFs within a fabric are necessary and may go all the way down to the leaves. How such tunnels are provisioned is outside the specification here but it will necessitate basically flat distribution of the loopbacks of the ToFs across whole fabric via e.g. redistribution of some RIFT routes in northbound and southbound direction or an equivalent scheme.

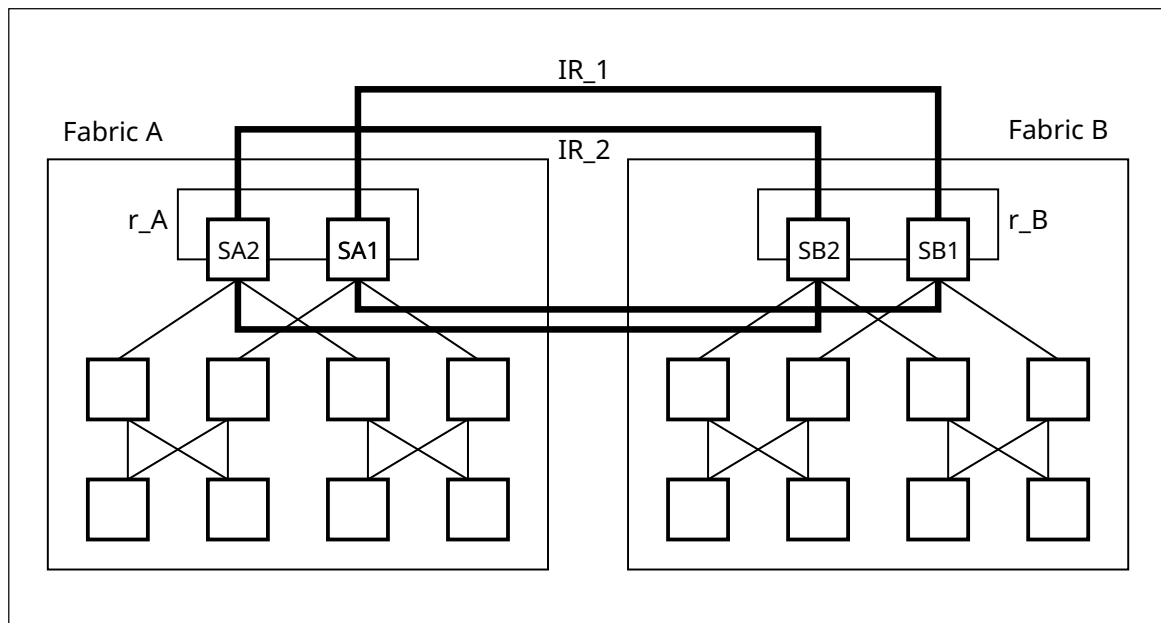


Figure 3: Multi-Plane CLOS Fabrics Connected in Sparse Dragonfly

5. Forwarding Considerations

Since RIFT is being extended with the concept of "indirect inter fabric next hop" and IP packets do not carry any marking as the path they have taken indiscriminate forwarding using non-shortest paths at ToF level may loop in inter fabric case. To prevent this the ToFs have to maintain the concept of a "split horizon" on the arriving traffic. Any traffic arriving at the ToF that is targeted at the prefix within its fabric can be forwarded without any further considerations. On the other hand, traffic arriving at a inter fabric link MUST use a FIB which does not contain the indirect inter fabric next hops and hence the FIB used to forward traffic on the IF inter faces MUST NOT include the results of indirect next hop computation. The solution will naturally limit any non-shortest inter fabric path in ToF case to maximally one alternate next hop. Observe that per inter face specific FIB is nothing particularly special, any technology supporting VPN or trunking today is already capable of provisioning inter face specific forwarding behavior.

6. Partitioning of inter Fabric Planes

A special case where a plane within a remote fabric breaks down is not noticeable in another fabric and hence the traffic can black hole since we do suppress the IF links during negative disaggregation normally. To detect the condition reliably a ToF has to compute the inter fabric view of all the other ToFs in its own fabric while including IF links and consider the resulting difference as "inter fabric negative disaggregation". This is possible but at scale can present significant computational load and is left therefore as optional behavior. Additionally, even when the fabric is a single plane fabric it must be then ringed at ToF level since otherwise the ToFs do not see the inter fabric planes they are not part of as an IF ring.

The same computation will deal with an even stranger case of a double failure on the IF links where a ToF becomes completely separated from the other fabrics. It will detect this and initiate negative disaggregation for the according prefixes.

7. Specification

Precise schema changes and computation algorithms are to be provided in future version of the draft in detail. Basically the LIEs and Node TIEs need to be extended by fabric_id and DF+ mode indication and computations described conceptually in former chapters tightly specified.

8. Summary Overview

A final [Figure 4](#) is provided to map things back to the usual dragonfly sparse topology and show the concepts in action.

We see three fabrics, each of them multi-plane (though mixes are absolutely possible as long the number of ToFs connected to dragonfly are kept the same). The fat links represent the "IF horizon", i.e. any traffic coming from those links cannot use alternate next hops to the

destination. In this example traffic from LA11 going through PA11 and SA2 towards LC11 is given two choices of next hops, either SC2 or SB2. Now that it entered the IF horizon in case SB2 receives it no further alternate next hops will be used but traffic will be handed off to SC2 which applies the same rule and in this case actually forwards the traffic into the fabric.

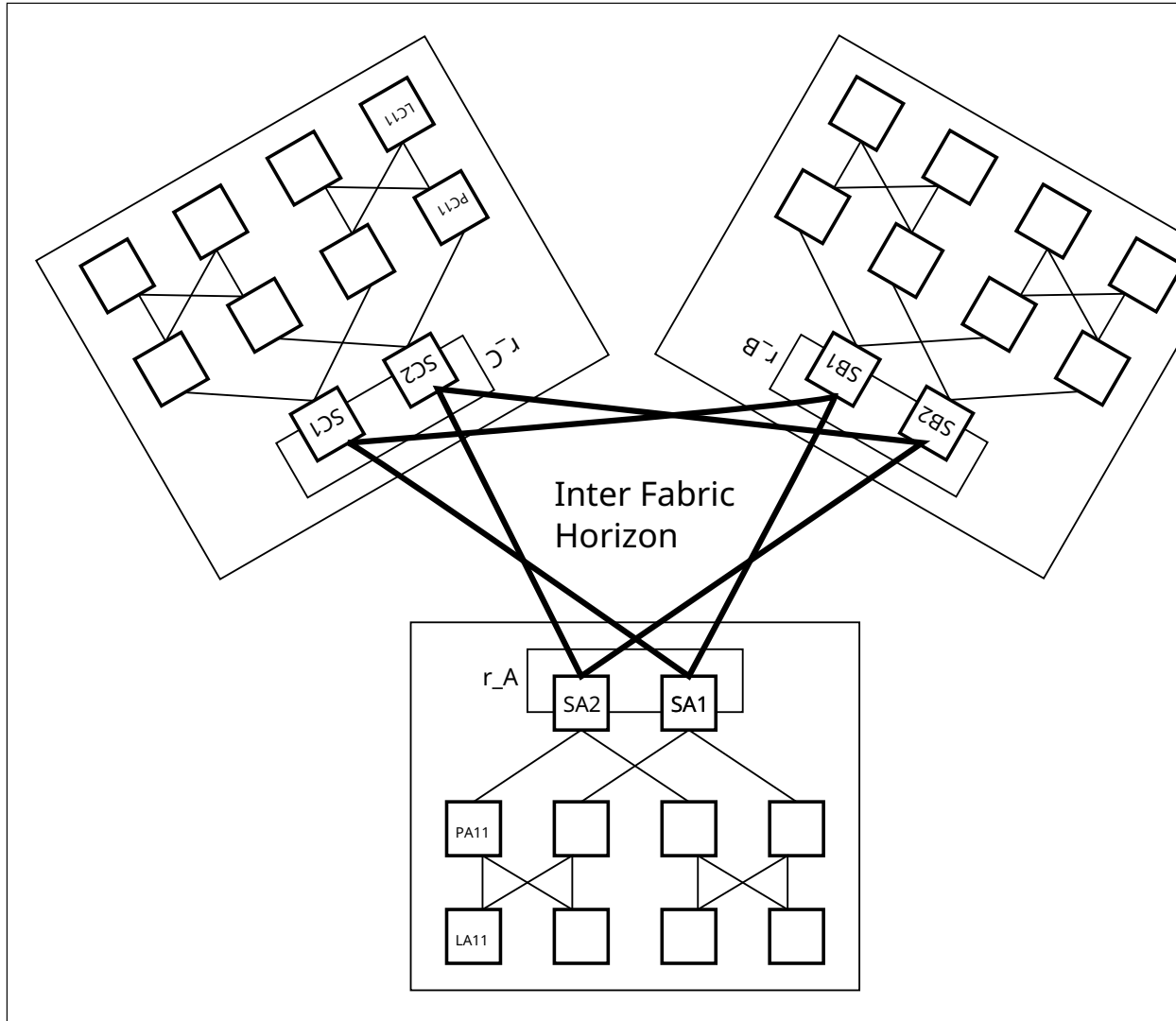


Figure 4: Multi-Plane CLOS Fabrics Connected in Sparse Dragonfly

9. IANA Considerations

This document requests allocation for the following RIFT codepoints.

TBD

10. Security Considerations

TBD

11. Acknowledgements

Dmitry Afanasiev's ideas around his work with BGP and dragonfly started interesting discussions, and he provided the crucial split horizon forwarding idea. Jeff Tantsura encouraged the work from its initial conception. Many thanks to Benson Muite for ASCII figures.

12. References

12.1. Informative References

12.2. Normative References

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